

HOLMES REEF, AustraliaWMO: **942890**Lat: **16.468S**Lon: **147.873E**Elev: **21**StdP: **14.68**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	70.5	71.5	N/A	N/A	N/A	N/A	N/A	N/A	32.0	73.6	30.5	73.9	18.8	130	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.0	86.1	N/A	85.4	N/A	84.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	10.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.4	29.3	27.5	68.5	88.4	2.6	1.1	66.7	89.2	65.2	89.9	63.7	90.5	61.9	91.3		
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.6	82.2	82.6	81.6	80.0	78.0	75.8	74.4	74.3	75.6	77.5	79.6	81.5		
	DBStd	3.33	1.46	1.58	1.72	1.43	1.57	1.63	1.61	1.42	1.44	1.46	1.40	1.55		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	10424	998	913	980	901	868	773	756	752	767	852	887	977		
	CDD65	4949	533	493	515	451	403	323	291	287	317	387	437	512		
	CDH74	42378	6251	5954	5902	4538	3120	1533	777	580	1236	2665	4102	5720		
(13)	CDH80	7741	1840	1965	1574	552	76	2	0	0	2	44	327	1359		
(14)	Wind	WSAvg	16.0	13.7	13.4	15.3	18.9	18.7	18.4	17.2	15.4	14.7	13.7	13.1		
(15) Precipitation	PrecAvg	39.3	6.3	7.5	8.3	3.9	2.3	1.6	1.2	0.9	0.5	1.0	1.5	4.1		
	PrecMax	61.9	14.1	16.9	28.7	11.5	9.1	4.6	5.4	4.1	1.7	9.3	9.5	15.7		
	PrecMin	12.9	0.1	1.3	0.4	0.4	0.4	0.1	0.1	0.2	0.0	0.0	0.0	0.2		
	PrecStd	11.5	4.2	4.3	5.6	3.5	2.2	1.1	1.2	0.9	0.4	1.7	2.1	3.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.8	87.3	86.6	83.6	82.4	80.4	78.9	78.3	80.2	82.5	84.2	86.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	85.6	86.2	85.4	82.8	81.6	79.3	78.1	77.4	79.2	81.2	83.0	85.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	84.9	85.5	84.6	82.6	80.9	78.7	77.4	76.8	78.3	80.3	82.3	84.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	84.4	84.7	84.1	82.1	80.3	78.1	76.9	76.2	77.5	79.6	81.7	83.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.9	3.0	3.0	2.2	2.1	2.3	2.3	1.9	1.7	1.9	2.1	2.4		
		MCDBR	3.6	3.6	3.6	2.8	2.7	2.6	2.8	2.4	2.4	2.7	2.9	3.2		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.392	0.392	0.400	0.388	0.376	0.378	0.354	0.359	0.377	0.390	0.392	0.390		
	taud		2.600	2.610	2.569	2.587	2.581	2.565	2.618	2.588	2.515	2.496	2.531	2.573		
	Ebn at Noon		302	299	290	283	275	268	279	287	292	296	300	302		
	Edh at Noon		33	33	33	30	29	28	27	30	34	36	35	34		
(44) All-Sky Solar Radiation	RadAvg		2078	1997	1800	1693	1470	1314	1413	1701	2041	2240	2310	2237		
	RadStd		186	171	137	133	126	101	88	105	70	102	173	239		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page